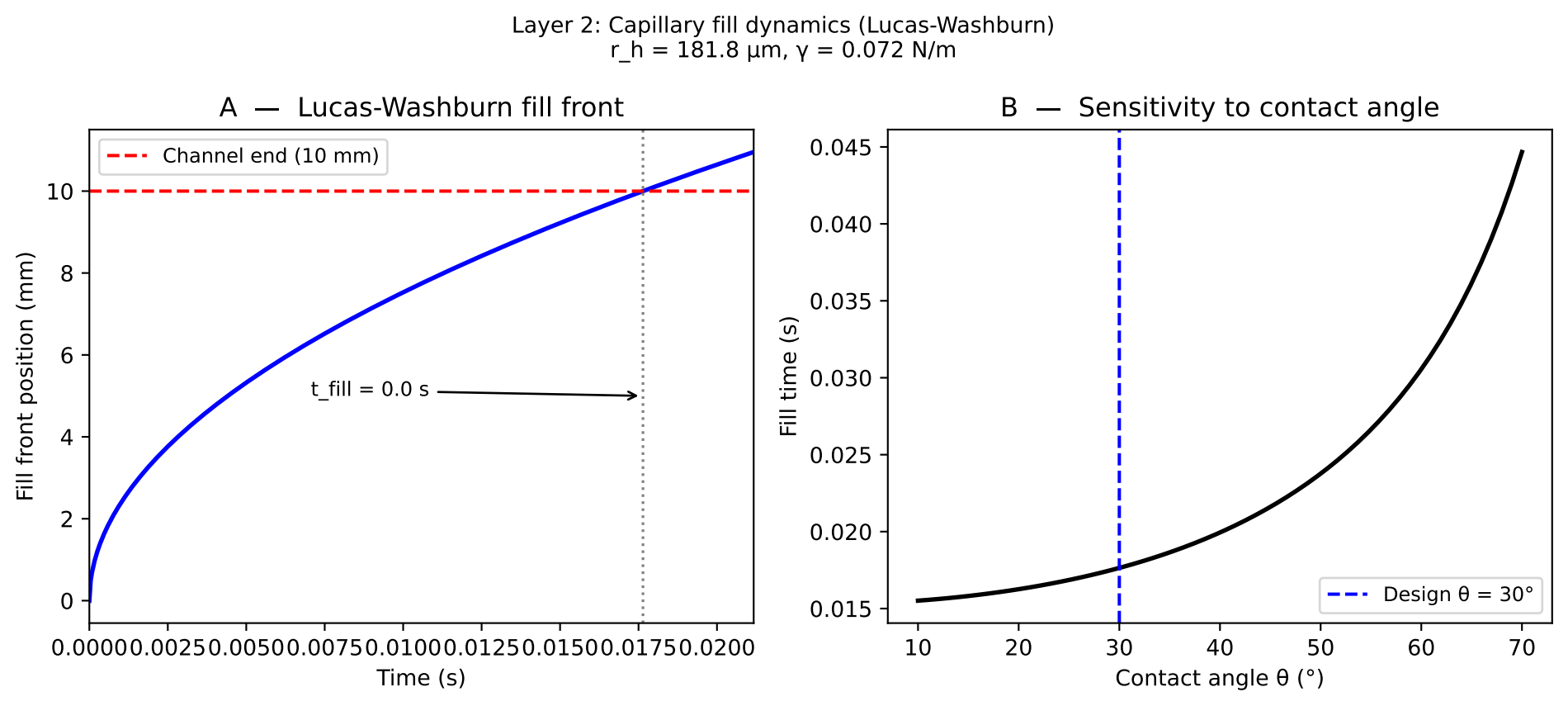
Layer 2: Capillary fill dynamics (Lucas-Washburn)
r_h = 181.8 µm, γ = 0.072 N/m
A — Lucas-Washburn fill front
Channel end (10 mm)
t_fill = 0.0 s
Fill front position (mm)
Time (s)
0.0000
0.0025
0.0050
0.0075
0.0100
0.0125
0.0150
0.0175
0.0200
0
2
4
6
8
10
B — Sensitivity to contact angle
Design θ = 30°
Fill time (s)
Contact angle θ (°)
0.015
0.020
0.025
0.030
0.035
0.040
0.045
10
20
30
40
50
60
70